CLASS IMBALANCE

- Assume a binary classifier diagnoses a serious medical condition.
- Label distribution is often **imbalanced**, i.e, not many people have the disease.
- Evaluating on mce is often inappropriate for scenarios with imbalanced labels:
  - Assume that only 0.5 % have the disease.
  - Always predicting "no disease" has an mce of 0.5 %, corresponding to very high accuracy.
  - This sends all sick patients home $\rightarrow$ bad system
- This problem is known as the **accuracy paradox**.

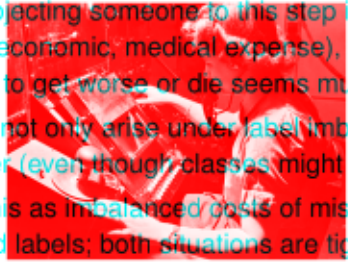
IMBALANCED COSTS

Classifying all observations as "no disease" (green) yields top accuracy simply because the "disease" occurs so rarely → accuracy paradox.

- Another point of view is **imbalanced costs**.
- In our example, classifying a sick patient as healthy should incur a much higher cost than classifying a healthy patient as sick.
- The costs depend a lot on what happens next: we can well assume that our system is some type of screening filter, and often the next step after labeling someone as sick might be a more invasive, expensive, but also more reliable test for the disease.
- Erroneously subjecting someone to this step is undesirable (psychological, economic, medical expense), but sending someone home to get worse or die seems much more so.
- Such situations not only arise under label imbalance, but also when costs differ (even though classes might be balanced).
- We could see this as imbalanced costs of misclassification, rather than imbalanced labels; both situations are tightly connected.

ROC ANALYSIS COSTS

- **ROC analysis** is a subfield of ML which studies the evaluation of binary prediction systems.
- ROC stands for "receiver operating characteristics" and was initially developed by electrical engineers and radar engineers during World War II for detecting enemy objects in battlefields – still has the funny name.
- In our example, classifying a sick patient as healthy should incur a much higher cost than classifying a healthy patient as sick.
- The costs depend a lot on what happens next. We can well assume that our system is some type of screening filter, and often the next step after labeling someone as sick might be a more invasive, expensive, but also more reliable test for the disease.
- Erroneously subjecting someone to this step is undesirable (psychological, economic, medical expense), but sending someone home to get worse or die seems much more so.
- Such situations not only arise under label imbalance, but also when costs differ (even though classes might be balanced).
- We could see this as imbalanced costs of misclassification, rather than imbalanced labels; both situations are tightly connected.



http://media.iwm.org.uk/iwm/mediaLib//39/media-39565/large.jpg

LABELS: ROC METRICS

From the confusion matrix (binary case), we can calculate "ROC" metrics.

		True Class y		
		$+$	$-$	
Pred. $\hat{y}$	$+$	TP	FP	$\rho_{PPV} = \frac{TP}{TP+FP}$
	$-$	FN	TN	$\rho_{NPV} = \frac{TN}{FN+TN}$
		$\rho_{TPR} = \frac{TP}{TP+FN}$	$\rho_{TNR} = \frac{TN}{FP+TN}$	$\rho_{ACC} = \frac{TP+TN}{TOTAL}$

- Problem: if we were able to specify costs precisely, we could evaluate or even optimize on them.
- This important subfield of ML is called **cost-sensitive learning**, which we will not cover in this lecture until
- Unfortunately, users find it notoriously hard to come up with precise cost figures in imbalanced scenarios.
- Evaluating "from different perspectives", with multiple metrics, often helps to get a first impression of system quality.

Imbalanced costs: classifying incorrectly as "no disease" incurs very high cost.

- True positive rate ρ_{TPR}: how many of the true 1s did we predict as 1?
- True Negative rate ρ_{TNR}: how many of the true 0s did we predict as 0?
- Positive predictive value ρ_{PPV}: if we predict 1, how likely is it a true 1?
- Negative predictive value ρ_{NPV}: if we predict 0, how likely is it a true 0?
- Accuracy ρ_{ACC}: how many instances did we predict correctly?

LABELS: ROC METRICS

Example:

- **ROC analysis** is a subfield of ML which studies the evaluation of binary prediction systems.
- ROC stands for "receiver operating characteristics" and was initially developed by electrical engineers and radar engineers during World War II for detecting enemy objects in battlefields still has the funny name.

		Actual Class y		
		Positive	Negative	
$\hat{y}$ **Pred.**	Positive	**True Positive** (TP) = 20	**False Positive** (FP) = 180	Positive predictive value = TP / (TP + FP) = 20 / (20 + 180) = **10%**
	Negative	**False Negative** (FN) = 10	**True Negative** (TN) = 1820	Negative predictive value = TN / (FN + TN) = 1820 / (10 + 1820) ≈ **99.5%**
		True Positive Rate = TP / (TP + FN) = 20 / (20 + 10) ≈ **67%**	True Negative Rate = TN / (FP + TN) = 1820 / (180 + 1820) = **91%**	

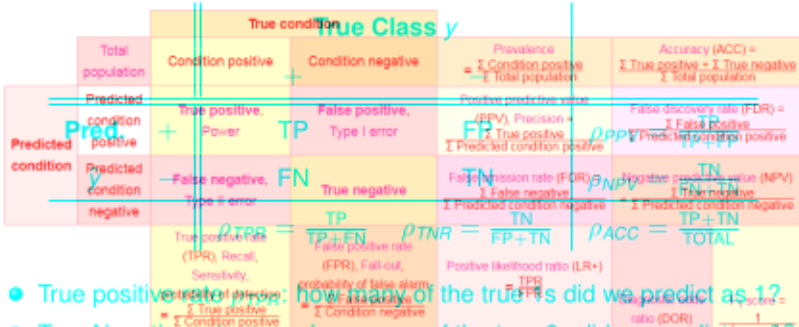
https://en.wikipedia.org/wiki/Receiver_operating_characteristic

http://media.iwm.org.uk/iwm/mediaLib//39/media-39665/large.jpg

MORE METRICS AND ALTERNATIVE TERMINOLOGY

From the confusion matrix (binary case), we can calculate "ROC"

Unfortunately, for many concepts in ROC, 2-3 different terms exist.



- True positive rate ρ_{TPR}: how many of the true 1s did we predict as 1?
- True Negative rate ρ_{TNR}: how many of the true 0s did we predict as 0?
- Positive predictive value ρ_{PPV}: if we predict 1, how likely is it a true 1?
- Negative predictive value ρ_{NPV}: if we predict 0, how likely is it a true 0?
- Accuracy ρ_{ACC}: how many instances did we predict correctly?

▸ Clickable version/picture source ▸ Interactive diagram

LABELS: F MEASURES

Example:

- It is difficult to achieve high **positive predictive value** and high **true positive rate** simultaneously.
- A classifier predicting more positive will be more sensitive (higher ρ_{TPR}), but it will also tend to give more *false* positives (lower ρ_{TNR}, lower ρ_{PPV}).
- A classifier that predicts more negatives will be more precise (higher ρ_{PPV}), but it will also produce more *false* negatives (lower ρ_{TPR}).

The F_1 **score** balances two conflicting goals:

1. Maximizing positive predictive value
2. Maximizing true positive rate

ρ_{F_1} is the harmonic mean of ρ_{PPV} and ρ_{TPR}:

$$\rho_{F_1} = 2 \cdot \frac{\rho_{PPV} \cdot \rho_{TPR}}{\rho_{PPV} + \rho_{TPR}}$$

https://en.wikipedia.org/wiki/Receiver_operating_characteristic

Note that this measure still does not account for the number of true negatives.

WHICH METRIC TO USE?

- As we have seen, there is a plethora of methods.
  - This leaves practitioners with the question of which to use.
- Consider a small benchmark study.
  - We let k-NN, logistic regression, a classification tree, and a random forest compete on classifying the `credit risk` data.
  - The data consist of 1000 observations of borrowers' financial situation and their creditworthiness (good/bad) as target.
  - Predicted probabilities are thresholded at 0.5 for the positive class.
  - Depending on the metric we use, learners are ranked differently according to performance (value of respective performance measure in parentheses):

metric	k-NN	logistic regression	random forest	CART
TPR	2 (0.8777)	3 (0.8647)	1 (0.9257)	4 (0.8357)
TNR	4 (0.3765)	2 (0.4737)	3 (0.4072)	1 (0.4911)
PPV	4 (0.7665)	1 (0.7947)	3 (0.7842)	2 (0.7925)
NPV	2 (0.8178)	2 (0.8275)	1 (0.8488)	4 (0.8130)
AUC	4 (0.7936)	2 (0.7731)	1 (0.7902)	3 (0.7293)
ACC	4 (0.7270)	2 (0.7490)	1 (0.7700)	3 (0.7320)

▸ Clickable version/picture source

LABELS: F_1 MEASURE

- It is difficult to achieve high **positive predictive value** and high **true positive rate** simultaneously.
- A classifier predicting more positive will be more sensitive (higher ρ_{TPR}), but it will also tend to give more *false* positives (lower ρ_{TNR}, lower ρ_{PPV}).
- A classifier that predicts more negatives will be more precise (higher ρ_{PPV}), but it will also produce more *false* negatives (lower ρ_{TPR}).

The F_1 **score** balances two conflicting goals:

1. Maximizing positive predictive value
2. Maximizing true positive rate

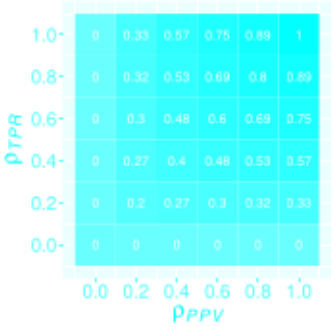
ρ_{F_1} is the harmonic mean of ρ_{PPV} and ρ_{TPR}:

$$\rho_{F_1} = 2 \cdot \frac{\rho_{PPV} \cdot \rho_{TPR}}{\rho_{PPV} + \rho_{TPR}}$$

Note that this measure still does not account for the number of true negatives.

LABELS: F_1 MEASURE

F_1 score for different combinations of ρ_{PPV} & ρ_{TPR}.
$\rightarrow$ Tends more towards the lower of the two combined values.



- A model with $\rho_{TPR} = 0$ (no positive instance predicted as positive) or $\rho_{PPV} = 0$ (no true positives among the predicted) has $\rho_{F_1} = 0$.
- Always predicting "negative": $\rho_{F_1} = 0$.
- Always predicting "positive":
  $\rho_{F_1} = 2 \cdot \rho_{PPV}/(\rho_{PPV} + 1) = 2 \cdot n_+/(n_+ + n)$,
  which will be small when the size of the positive class n_+ is small.

WHICH METRIC TO USE?

- As we have seen, there is a plethora of methods.
  → This leaves practitioners with the question of which to use.
- Consider a small benchmark study.
  - We let k-NN, logistic regression, a classification tree, and a random forest compete on classifying the `credit risk` data.
  - The data consist of 1000 observations of borrowers' financial situation and their creditworthiness (good/bad) as target.
  - Predicted probabilities are thresholded at 0.5 for the positive class.
  - Depending on the metric we use, learners are ranked differently according to performance (value of respective performance measure in parentheses):

metric	k-NN	logistic regression	random forest	CART
TPR	2 (0.8777)	3 (0.8647)	1 (0.9257)	4 (0.8357)
TNR	4 (0.3764)	2 (0.4797)	3 (0.4072)	1 (0.4911)
PPV	4 (0.7965)	1 (0.7947)	3 (0.7842)	2 (0.7925)
F1	3 (0.8179)	2 (0.8279)	1 (0.8488)	4 (0.8150)
AUC	4 (0.7092)	2 (0.7731)	1 (0.7902)	3 (0.7293)
ACC	4 (0.7270)	2 (0.7490)	1 (0.7700)	3 (0.7320)

WHICH METRIC TO USE?

- We need not expect overly large discrepancies in general, but neither will we always see an unambiguous picture.
- Different metrics emphasize different aspects of performance.
  $\rightarrow$ The choice should be made in the domain context.
- For practitioners it is vital to understand what should be evaluated exactly, and which measure is appropriate.
  - Regarding credit risk, for instance, defaults are to be avoided, but not at all cost.
  - The bank must undertake a certain risk to remain profitable, so a more balanced measure such as the F_1 score might be in order.
  - On the other hand, a system detecting weapons at an airport should be able to achieve very high true positive rates, even if this comes at the expense of some false alarms.